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Yuanyuan Chen

1,258 citations · h-index 11 · Oct. 2026

Education

2021 – 2026 **PhD in Computer Science**, *Nanyang Technological University*, Singapore
Advisor: Assoc. Prof. Han Yu

2005 – 2009 **BSc in Mathematics and Applied Mathematics**, *Fuzhou University*, China

Research Interests

- Trustworthy Federated Learning
- Explainable AI and Data Valuation
- Federated Graph Learning
- Large Language Models and Generative AI

Work Experience

Aug. 2026 – present **Research Fellow**, *Bake AI Inc*, USA
Provide independent research and technical advisory services on research infrastructure, software development, and systems architecture. Review designs and code for reliability, performance, and security; develop agreed prototypes, research tools, and technical documentation.

Apr. 2026 – Jun. 2026 **Founding CTO**, *Innoxsea Pte Ltd*, Singapore
Built a knowledge graph-based life sciences information database and reasoning engine; led technical strategy and system architecture.

Dec. 2025 – Mar. 2026 **Research Fellow**, *Bake AI Inc*, USA
Led development of crust (**440+ GitHub stars**), an open-source security framework that intercepts and blocks dangerous AI agent actions in real time. Contributed to Visual Aesthetic Benchmark (VAB), evaluating frontier multimodal models' aesthetic judgment.

Jul. 2025 – Nov. 2025 **Research Assistant**, *Deeper Semantics Lab, NTU*, Singapore

Oct. 2019 – Jun. 2025 **Research Assistant**, *NTU–WeBank Research Centre on FinTech*, Singapore
Researched efficient federated learning, healthcare applications, graph learning, and LLM prompt optimization. FedOBD reduced training communication overhead by over 70%; CAreFL supported collaborative modeling across eight medical institutions.

Jun. 2019 – Sep. 2019 **Research Intern**, *WeBank*, Shenzhen, China
Implemented a federated visual object detection system for privacy-preserving safety monitoring.

May 2016 – Feb. 2019 **System Architect**, *DeepIR Inc*, Xiamen, China
Handled system architecture, model integration, inference services, and deployment for AI safety monitoring at Beijing 2022 Winter Olympics venue construction sites and license plate recognition at the Hong Kong Customs checkpoint of the Hong Kong–Zhuhai–Macau Bridge.

Jul. 2009 – Apr. 2016 **Senior Software Engineer**, *ChinaSource Ltd*, Xiamen, China
Developed a commercial email framework using open-source components, supporting POP3, IMAP, and SMTP.

Open Source Contributions

2024–present **SIG-archinfra Approver**, *ONNX*
SIG-archinfra Approver.

2020–present **Outside Collaborator**, *PyTorch*
Outside-collaborator status with code-review and issue-triage rights on core repositories. Sustained contributions since 2020 spanning the PyTorch ecosystem (**xla**, **vision**, **functorch**, **executorch**, **gloo**, **builder**, **cpuinfo**, **text**) were recognised with the **PyTorch Community Award** (2023 Newcomer Nominee).

2024–present **Contributor**, *NVIDIA NVFlare*
Upstreamed own **FedOBD** (IJCAI 2023) and its derived **ADAQUANT** quantization scheme into the NVFlare research dir during NVIDIA FLARE Summer of Code 2025 (PR #3666).

2023–present **Contributor**, *Hugging Face*
Sustained contributor across the Hugging Face ecosystem — **transformers**, **accelerate**, **trl**, **peft**, **datasets** — with the bulk of contributions landing in **transformers**.

Publications

Conference Proceedings

- Yichen Feng, Yuetai Li, Chunjiang Liu, **Yuanyuan Chen**, Fengqing Jiang, Yue Huang, Hang Hua, et al., "Visual Aesthetic Benchmark: Can Frontier Models Judge Beauty?," *COLM* 2026 · Final PDF in pre-proceedings; DOI pending.
- Yichen Feng, Yuetai Li, Chunjiang Liu, **Yuanyuan Chen**, Fengqing Jiang, Yue Huang, Hang Hua, et al., "Visual Aesthetic Benchmark: Can Frontier Models Judge Beauty?," *ICML* 2026 Workshop on Human-AI Co-Creativity · Camera-ready report (non-archival).
- Minghui Chen, Ruinan Jin, Wenlong Deng, **Yuanyuan Chen**, Zhi Huang, Han Yu, and Xiaoxiao Li, "Can Textual Gradient Work in Federated Learning?," *ICLR* 2025 · Proceedings.
- Chen Chen, Ruizhe Li, Yuchen Hu, **Yuanyuan Chen**, Chengwei Qin, and Qiang Zhang, "Overcoming Catastrophic Forgetting by Exemplar Selection in Task-oriented Dialogue System," *ACL* 2024.
- Anran Li*, **Yuanyuan Chen***, Jian Zhang, Mingfei Cheng, Yihao Huang, Yueming Wu, Anh Tuan Luu, and Han Yu, "Historical Embedding-Guided Efficient Large-Scale Federated Graph Learning," *SIGMOD* 2024.
- **Yuanyuan Chen***, Zichen Chen*, Pengcheng Wu, and Han Yu, "FedOBD: Opportunistic Block Dropout for Efficiently Training Large-scale Neural Networks through Federated Learning," *IJCAI* 2023.
- **Yuanyuan Chen***, Zichen Chen*, Sheng Guo, Yansong Zhao, Zelei Liu, Pengcheng Wu, Chengyi Yang, Zengxiang Li, and Han Yu, "Efficient Training of Large-scale Industrial Fault Diagnostic Models through Federated Opportunistic Block Dropout," *IAAI* 2023.
- Zelei Liu*, **Yuanyuan Chen***, Yansong Zhao, Han Yu, Yang Liu, Renyi Bao, Jinpeng Jiang, Zaiqing Nie, Qian Xu, and Qiang Yang, "Contribution-Aware Federated Learning for Smart Healthcare," *IAAI* 2022.
- **Yuanyuan Chen**, Boyang Li, Han Yu, Pengcheng Wu, and Chunyan Miao, "HyDRA: Hypergradient Data Relevance Analysis for Interpreting Deep Neural Networks," *AAAI* 2021.
- Yang Liu, Anbu Huang, Yun Luo, He Huang, Youzhi Liu, **Yuanyuan Chen**, Lican Feng, Tianjian Chen, Han Yu, and Qiang Yang, "FedVision: An Online Visual Object Detection Platform Powered by Federated Learning," *IAAI* 2020.

Journal Articles

- Anran Li, **Yuanyuan Chen**, Wenjun Long, Yu Yin, Yan Hu, Hyunjae Kim, Weipeng Zhou, et al. 2026. "Toward Federated Large Language Models in Medicine: A Parameter-Efficient Framework for Privacy-Preserving, Multi-Institutional Adaptation." *npj Digital Medicine* (August).

Article in Press (publisher's accepted manuscript)

- Chuan Sun, **Yuanyuan Chen**, Xiaoli Tang, and Han Yu. 2026. "Efficient Federated Learning With Mean Block Difference-Based Global Aggregation and Patience-Based Local Training." *IEEE Transactions on Knowledge and Data Engineering* 38, no. 7 (July): 4270–4286.

- Rui Liu, **Yuanyuan Chen**, Anran Li, Yi Ding, Han Yu, and Cuntai Guan. 2024. "Aggregating intrinsic information to enhance BCI performance through federated learning." *Neural Networks* 172:106100.
- Zelei Liu, **Yuanyuan Chen**, Yansong Zhao, Han Yu, Yang Liu, Renyi Bao, Jinpeng Jiang, Zaiqing Nie, Qian Xu, and Qiang Yang. 2023. "CAreFL: Enhancing smart healthcare with Contribution-Aware Federated Learning." *AI Magazine* 44 (1): 4–15.
- Hao Wang, Boyang Li, Haoming Zhong, Ahong Xu, Yingjie Huang, Jingfu Zou, **Yuanyuan Chen**, et al. 2022. "Smart Decision-Support System for Pig Farming." *Drones* 6 (12).
- Zelei Liu, **Yuanyuan Chen**, Han Yu, Yang Liu, and Lizhen Cui. 2022. "GTG-Shapley: Efficient and Accurate Participant Contribution Evaluation in Federated Learning." *ACM Transactions on Intelligent Systems and Technology* 13, no. 4 (May).
- Yang Liu, Anbu Huang, Yun Luo, He Huang, Youzhi Liu, **Yuanyuan Chen**, Lican Feng, Tianjian Chen, Han Yu, and Qiang Yang. 2021. "Federated Learning-Powered Visual Object Detection for Safety Monitoring." *AI Magazine* 42, no. 2 (October): 19–27.

Selected Bug Reports & Fixes

2025–2026 **Approver**, *ONNX*
As a SIG-archinfra Approver, reported and fixed security issues in model validation and loading, including recursive local-function cycles, external-data TOCTOU, raw-data heap overflow, and invalid graph outputs (PRs #7815, #7790, #8109, #8121).

2021 **Bug report**, *NVIDIA DALI / nvJPEG*
Reproduced GPU memory growth during repeated iterator creation and teardown; the report led to a DALI fix and an nvJPEG fix in CUDA 11.5 Update 1 (DALI PR #3416).

2015 **Bug report**, *GMime*
Diagnosed a 28-byte memory leak in decode_route with Valgrind; fixed upstream in GMime 2.6.21.

Projects

Award Winning Projects

2023 **FedOBD**
Reduced federated training communication overhead through opportunistic transmission of quantized model blocks; deployed for industrial fault diagnosis at ENN Group.

2022 **Contribution-Aware Federated Learning Framework**
Enabled privacy-preserving healthcare collaboration through efficient participant contribution evaluation and contribution-aware model aggregation.

2020 **FedVision**
Built a platform for federated computer vision, enabling collaborative object detection without sharing raw video data.

Awards & Honors

Academic Awards

2024 **Certificate of Accomplishment** for your outstanding performance and dedication towards the ACM, Siggraph & Sigmobile NTU Student Chapter, *NTU Association for Computing Machinery, Siggraph and Sigmobile*, Singapore

2023 **Deployed Application Award** for "Efficient Training of Large-scale Industrial Fault Diagnostic Models through Federated Opportunistic Block Dropout", *Association for the Advancement of Artificial Intelligence*

2022 **Innovative Application Award** for "Contribution-Aware Federated Learning for Smart Healthcare", *Association for the Advancement of Artificial Intelligence*

2020 **Innovative Applications of Artificial Intelligence Award** for "FedVision: An Online Visual Object Detection Platform Powered by Federated Learning", *Association for the Advancement of Artificial Intelligence*

Industry Honors

Apr. 2026 **GitHub Secure Open Source Fund**, *GitHub*

Competitively selected for sustained security work on **ONNX**, the open standard for machine learning model representation.

2026 **OpenAI Patch the Planet**, *OpenAI, with Trail of Bits*

Participated with **ONNX** in an AI-assisted open-source security initiative pairing OpenAI's cyber-capable models with expert human review to find and land vulnerability fixes. ONNX's selection was confirmed in its official August 2026 meeting minutes.

2025 **NVIDIA FLARE Summer of Code 2025**, *NVIDIA FLARE*

2023 **PyTorch Community Awards** for PyTorch Newcomer Nominee, *The Linux Foundation*

Recognition for contributions to **PyTorch**, the world's leading open-source deep learning framework powering modern AI research and industry deployments.

2021 **Mars 2020 Helicopter Contributor**, *GitHub*

2020 **Arctic Code Vault Contributor**, *GitHub*

University Awards

2024 **5 Years Long Service Award** in appreciation for the service rendered to the university, *Nanyang Technological University*, Singapore

2022 **Top 5 Most Popular Video Award** for Trust-based Open Collaborative Federated Learning, *College of Engineering, Nanyang Technological University*, Singapore

Community Awards

2025 **Certificate of Distinguished Service** in appreciation of your valuable contributions to Pei Chun Alumni Association, *Pei Chun Alumni Association*, Singapore

2025 **Certificate of Appreciation** for your contributions to the successful launch of SG 60 App-driven Belanja A Meal (BAM) Initiative, *People's Association*, Singapore

2025 **Certificate of Appreciation** for your contributions to the initiation and innovative digitalisation of Merchant Dollar Deals @ Woodlands, *People's Association*, Singapore

2025 **Certificate of Appreciation** for your generous contribution to Woodlands Community Club Building Fund, *People's Association*, Singapore

2025 **Certificate of Distinction** for exemplary performance as Founding Member of the Woodlands Digital Office and Outstanding Contributions to the Community in 2025, *People's Association*, Singapore

2024 **Certificate of Appreciation** in appreciation for your kind contribution towards Yuhua Charity Walk 2024, *People's Association*, Singapore

2024 **Certificate of Distinction** for exemplary performance as Founding Member of the Woodlands Digital Office and Outstanding Contributions to the Community in 2024, *People's Association*, Singapore

2024 **Community Leadership Award** for outstanding leadership and dedication to our community for the year 2024, *People's Association*, Singapore

2023 **Certificate of Excellence** in recognition of outstanding contribution to our community for the year 2023, *People's Association*, Singapore

2023 **Community Leadership Award** for outstanding leadership and dedication to our community for the year 2023, *People's Association*, Singapore

2022 **Achiever's Award** for your exceptional contributions to promoting and persevering cultural heritage in Singapore, *People's Association*, Singapore

2022 **Certificate of Appreciation** for valuable contributions to the community in the year 2022, *People's Association*, Singapore

2022 **Certificate of Appreciation** in recognition of outstanding contribution to our community for the year 2022, *People's Association*, Singapore

2021 **Certificate of Appreciation** in recognition of outstanding contribution to our community for the year 2021, *People's Association*, Singapore

2020–2021 **Certificate of Appreciation** for valuable contributions to the Community from 2020-2021, *People's Association*, Singapore

2019–2020 **Certificate of Achievement** as a valued member of Woodlands CCC New Media Committee and Outstanding Contributions to the community from 2019-2020, *People's Association*, Singapore

Research Grants

2022–2025 **Participant, "Towards Real-time, Seamless, and Ubiquitous Edge Intelligence-empowered Metaverse"**, *Future Communications Research & Development Programme, IMDA Singapore (No. FCP-NTU-RG-2021-014)*, S$499,999.50

2021–2025 **Participant, "TrustFUL: Trustworthy Federated Ubiquitous Learning"**, *AI Singapore Programme, National Research Foundation Singapore (No. AISG2-RP-2020-019)*, S$6,883,200

2020–2024 **Participant, "NOURISH: Next-Generation Brain-Computer-Brain Platform"**, *RIE 2020 Advanced Manufacturing & Engineering Programmatic Fund, A*STAR Singapore (No. A20G8b0102)*, S$9,950,280

2019–2021 **Participant, "An End-to-end Adaptive AI-Assisted 3H Care (A3C) System"**, *AI Singapore Grand Challenge, National Research Foundation Singapore (No. AISG-GC-2019-003)*, S$4,600,000

Professional Services

Program Committee

2026 **The Fortieth AAAI Conference on Artificial Intelligence**, *AAAI-26*, Singapore

Academic Reviewer

2025 IEEE Transactions on Neural Networks and Learning Systems

2024 IEEE Transactions on Neural Networks and Learning Systems

2024 International Journal of Intelligent Systems

2023 ACM Transactions on Software Engineering and Methodology

2022 IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences

2022 IEEE Transactions on Neural Networks and Learning Systems

2020 The Thirty-Fourth AAAI Conference on Artificial Intelligence

Academic Conference Volunteer

2024 **The 2024 ACM Web Conference**, *WWW'24*, Singapore

2023 **The Thirty-Second International Joint Conference on Artificial Intelligence**, *IJCAI 2023*, Macao S.A.R.

Invited Talks & Presentations

2025 **NVIDIA FLARE Day APAC 2025**, *NVIDIA FLARE*, Online

2023 **Singapore ACM SIGKDD Symposium 2023**, *Singapore ACM SIGKDD Chapter*, Singapore

2018 **Invited Talk for Google Developer Groups**, *DevFest*, Xiamen, China

Membership

Professional Societies

2023–present **Member**, *Institute of Electrical and Electronics Engineers*

2022–present **Member**, *Association for Computing Machinery*

2022–present **Member**, *Association for the Advancement of Artificial Intelligence*